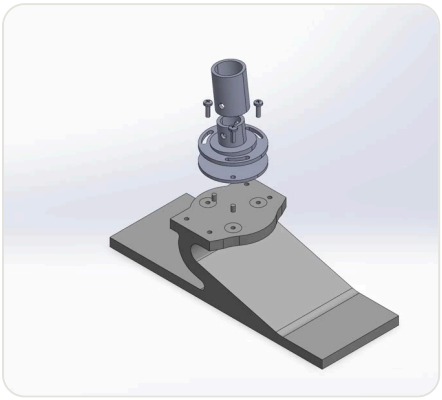
MECHANICAL ENGINEERING — UNIVERSITY OF LEEDS

Daniel Frankl

Final-year Mechanical Engineering student (1st Class) with hands-on CAD and FEA-based design experience spanning prosthetics, renewable energy systems and mechatronics. Proven track record delivering competition-winning designs (1st out of 33 teams) and conducting research at a leading biomechanics lab (Technion).

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MEDICAL DEVICE

Modular quick-connect prosthetic foot interface

eNABLe Lab, Technion — Summer 2026 research placement

3-DOF mechanism FEA-VALIDATED

Designed and FEA-validated a modular quick-connect interface for a prosthetic foot, enabling rapid tool-free swaps between foot components.

ROLE Sole design engineer on the mechanism, working within an established research lab methodology.

APPROACH Built a 3-degree-of-freedom connector mechanism in SolidWorks, then validated it under simulated gait loading using Winter's inverse dynamics method to derive realistic joint forces across the gait cycle. Ran static and fatigue FEA in SolidWorks Simulation against ISO 10328 and ISO 24562 structural test requirements for prosthetic components.

OUTCOME Mechanism passed simulated ISO loading cases with margin, with a full validation report suitable for onward physical prototyping.

KEY SKILLS SolidWorks · SolidWorks Simulation (FEA) · GD&T · ISO 10328 · ISO 24562 · Winter's inverse dynamics

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MECHANICAL DESIGN

Micro-hydroelectric turbine

University of Leeds design competition — ranked 1st of 33 teams

1st / 33 teams COMPETITION RANKING

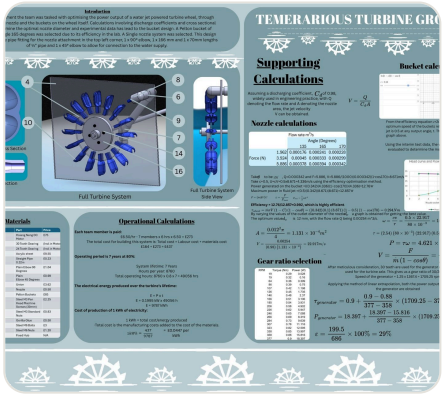
Designed a small-scale hydroelectric turbine for a low head-height site, optimised for manufacturability and power output within a fixed budget and material set.

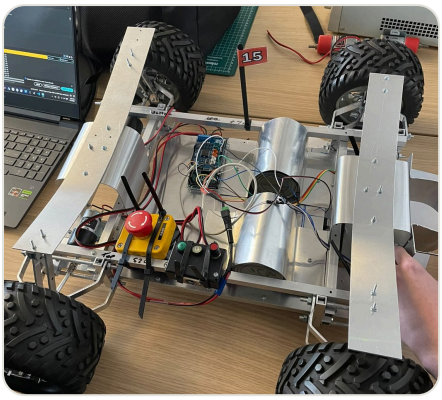
ROLE Design lead within a small team, responsible for the runner geometry and structural validation.

APPROACH Iterated blade and runner geometry using hand calculations and CAD-based flow reasoning, then validated the housing and shaft under load using FEA. Manufactured components using 3D printing and laser-cut acrylic.

OUTCOME Highest power output-to-cost ratio of all 33 competing teams, placing the design first overall.

KEY SKILLS SolidWorks · FEA · 3D printing · Acrylic fabrication · Manufacturing for cost





CONTROL SYSTEMS

The Daring Dash — autonomous buggy

University of Leeds team project

Payload delivered TARGET REACHED

Built an autonomous buggy capable of navigating a fixed course without manual control, combining custom chassis design with Arduino-based sensing and control.

ROLE Mechanical design and build, working alongside teammates on the control logic.

APPROACH Machined and assembled the chassis and drivetrain, then integrated ultrasonic and line-following sensors with an Arduino controller to drive navigation decisions.

OUTCOME On its competition run, the buggy navigated the course autonomously and deposited its payload in the target zone, meeting the competition's timing and reliability requirements.

KEY SKILLS SolidWorks · Arduino · CNC machining · Sensor integration

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EMBEDDED SOFTWARE

Drone flight control in C++

Independent project

1st in project RANKING

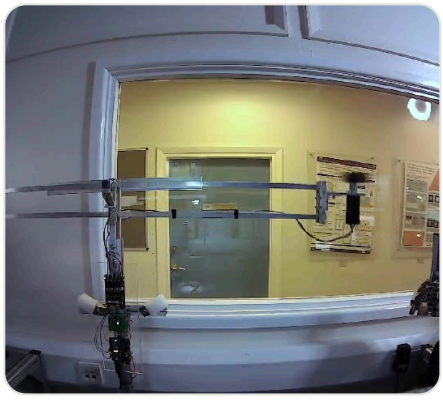
Programmed an Arduino-based drone controller in C++ to maintain a set flight height for a set duration.

ROLE Sole developer.

APPROACH Implemented PI control logic in C++ to hold a target flight height over a fixed duration, tuning gains through iterative testing on hardware.

OUTCOME Adjusted the code based on output data (RPM, drone weight) to bring the average error down to zero, achieving 1st in the project.

KEY SKILLS C++ · Arduino · PI control · Sensor fusion





DATA SCIENCE

ASA DataFest — Best Data Visualisation

University of Leeds Mathematics Department hackathon

Best Data Visualisation AWARD WON

Won Best Data Visualisation at the ASA DataFest hackathon, analysing a large real-world dataset against the clock as part of a 4-person team.

ROLE Worked within a team of four, taking ownership of a subset of the analysis and contributing to the final presentation design.

APPROACH The team isolated three core problems within a large, messy real-world dataset and split the workload accordingly. Cleaned and manipulated large CSV files in Excel, applied machine learning models and statistical correlation tests in Python to surface faults and relationships in the data, then built a PowerPoint presentation to communicate the findings clearly to a panel of judges.

OUTCOME Awarded Best Data Visualisation from the judging panel for presenting complex findings in a clear, visual, and easy-to-understand way.

KEY SKILLS Python · Machine learning · Statistical correlation analysis · Excel · Data cleaning · PowerPoint